Simple AI Attendance Web App

Project Technology Stack Specification

BACKEND & ARCHITECTURE	**Python 3** — Core programming language **Flask** — Web application framework & REST API server **Gunicorn** — WSGI HTTP production server **Jinja2** — Server-side HTML template rendering engine **Requests** — HTTP networking client for external APIs
DATABASE & STORAGE	**MongoDB** — NoSQL document database **PyMongo** — Python native MongoDB driver
FRONTEND & UI	**HTML5 & CSS3** — Semantic markup and responsive styling **JavaScript (Vanilla ES6+)** — Client-side dynamic interactions & AJAX **FontAwesome 6** — Vector icons and visual UI symbols
ARTIFICIAL INTELLIGENCE & BIOMETRICS	**OpenCV** — Real-time face detection & image preprocessing **NumPy** — High-performance vector and embedding calculations **Google Gemini AI (1.5 Flash)** — Autonomous field worker detour & anomaly auditor
MAPS & GEOLOCATION	**Leaflet.js** — Mobile-friendly interactive map library **OpenStreetMap** — Open map tile layer provider **OpenStreetMap Nominatim** — Geocoding and place name search engine **Komoot Photon API** — Secondary high-speed fallback geocoder **HTML5 Geolocation API** — Device GPS hardware coordinate acquisition
HARDWARE & MEDIA APIS	**WebRTC / MediaDevices API** — Live browser webcam video stream capture **Web Audio API** — Client-side synthetic dual-tone chime generation
DOCUMENT & REPORT GENERATION	**ReportLab** — Programmatic PDF generation (Payslips, Reports) **OpenPyXL** — Excel spreadsheet generation and export **Pillow (PIL)** — Image manipulation, face cropping & compression